Problem Description (Tech Interview)

(1) You are part of the core engineering team building an AI-assisted video management platform for smart warehouses. Security cameras stream activity metadata as a continuous chronological sequence of event tags, where each character represents a classified activity detected in a video frame:

- `'P'` : Person / Worker detected
- `'V'` : Forklift / Vehicle movement
- `'B'` : Unattended baggage or cargo container
- `'G'` : Security guard checkpoint scan
- `'D'` : Automated bay door opened/closed
- `'F'` : Hazard / Smoke indicator

During audit investigations, a safety officer specifies an **Incident Evidence Pattern** (T) containing critical event codes that must all occur together to confirm an incident.

Given two strings:

1. `video_stream` (S): A string representing the sequential chronological event frames recorded by the CCTV system.
2. `evidence_pattern` (T): A string representing the exact set and multiplicity of events required to validate the incident.

Your objective is to find the **shortest continuous clip (minimum length substring)** within `video_stream` that contains all the event codes present in `evidence_pattern`, including duplicates.

Resolution Rules:

- If multiple valid clips have the same minimum duration, return the clip that appears earliest in the timeline (smallest starting index).
- If no contiguous segment contains all required events, return an empty string `""` (or print `"Incident Incomplete"`).

Constraints

- 1 <= length(video_stream) <=2 *10^5
- 1 <= length(evidence_pattern) <= 10^4
- Both strings consist solely of uppercase and lowercase English letters (`A-Z`, `a-z`) and digits (`0-9`).
- **Time Complexity Target:** Strict O(N) where N = length(video_stream). Brute force solutions with O(N^2)or higher will fail execution timeouts.
- **Auxiliary Space Complexity Target:** O(1) auxiliary memory (only fixed-size frequency tables of size 128 or 256 are permitted; dynamic hash maps or memory allocations proportional to N are prohibited).

Example 1

Input:

video_stream = "PVDGDBPVG"

evidence_pattern = "PVB"

Output:

Clip: "BPV"

Start Frame: 5

End Frame: 7

Duration: 3

Explanation:

- The earlier sequence `"PVDGDB"` (frames 0 to 5) covers `'P'`, `'V'`, and `'B'`, but its length is 6 frames.
- The subsequence starting at index 5 and ending at index 7 is `"BPV"`. It contains `'B'`, `'P'`, and `'V'`, and its length is only 3 frames, which is the shortest possible window.

Example 2

Input:

video_stream = "GGDDVV"

evidence_pattern = "PV"

Output:

Incident Incomplete

Explanation: Event `'P'` never occurs in the stream, so the checklist cannot be fulfilled.

Example 3

Input:

video_stream = "PGPDPDVPB"

evidence_pattern = "PPV"

Output:

Clip: "PDVP"

Start Frame: 4

End Frame: 7

Duration: 4

Explanation: The pattern demands two instances of `'P'` and one instance of `'V'`. The segment from index 4 to 7 (`"PDVP"`) satisfies all counts with a minimal span of 4 frames.